THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Proportional Reasoning-1** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 8 : 12 : 21	Make the middle terms equal using their LCM
2	**(d)** 24 years	Let the ages be $3x$ and $2x$; solving $(3x+4)\times 5=(2x+4)\times 7$ gives $x=8$
3	**(b)** ₹ 600	Convert the reciprocals to whole-number ratio using the LCM
4	**(a)** 9 days	Find the work left in man-days and divide by the new number of men
5	**(c)** 5 : 4	Time is inversely proportional to speed
6	**(d)** 32	Adding the denominators keeps the ratio
7	**(a)** 4 : 9	Area $=$ side^2
8	**(b)** 27	$x=\frac{9\times 12}{4}$
9	**(a)** Both A and R are true and R is the correct explanation of A.	$3\times 12=36=4\times 9$, so the ratios are equal; R is the test.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Dividing both terms by 5 gives $3:5$ so A is true; R is also true but does not explain the simplification.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Perimeters $3:7$ (1); areas $9:49$ (1).	1+1
12	Take $x=7k$, $y=3k$ (1); $(10k):(4k)=5:2$ (1).	1+1
13	Ratio $\frac{1}{2}:\frac{1}{3}=3:2$ (1); parts ₹ 720 and ₹ 480 (1).	1+1
14	Perimeters $4:5$ (1); areas $16:25$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let ages be $3x$ and x (1); $(3x+4)\times 3=(x+4)\times 7$ gives $x=8$ (1); ages 24 and 8 years (1).	1+1+1
16	Total work $=360$ man-days; done $=120$ (1); left $=240$ (1); days $=240\div 20=12$ (1).	1+1+1
17	LCM of 2, 3, 4 is 12; ratio $6:4:3$ (2); parts ₹ 1200, ₹ 800, ₹ 600 (1).	2+1
	OR: Let ages be $5x$ and $2x$ (1); $(5x+6)\times 1=(2x+6)\times 2$ gives $x=6$ (1); ages 30 and 12 years (1).	1+1+1
18	Total work $=210$ man-days; done $=60$ (1); left $=150$ (1); days $=150\div 15=10$ (1).	1+1+1
	OR: LCM of 2, 4, 5 is 20; ratio $10:5:4$ (2); parts ₹ 2000, ₹ 1000, ₹ 800 (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{3}{3+2}=\frac{3}{5}$ (1). (ii) $\frac{3}{5}\times 10=6$ L (1). (iii) $10-6=4$ L (1). (iv) Curd $=\frac{3}{5}\times 25=15$ L and water $=10$ L (1); $15:10=3:2$ (1).	1+1+1+2
20	Investment $\times$ time: 12, 12, 8 (1); ratio $3:3:2$ (1); one part ₹ 150 (1); shares ₹ 450, ₹ 450, ₹ 300 (2).	1+1+1+2
	OR: Milk 50 L, water 40 L (1); in 18 L removed: milk 10 L, water 8 L (1); milk left 40 L (1); water now $40-8+18=50$ L (1); ratio $4:5$ (1).	1+1+1+1+1

Mistakes to avoid:

- Forming a ratio without converting units: 50 cm : 2 m is $50:200=1:4$, not $25:1$.
- Writing the ratio in the wrong order: the ratio of A to B is not the same as B to A.
- Dividing a total in the ratio $3:5$ by dividing it by 3 and 5 separately; first add the parts ($3+5=8$) and divide by 8.
- Adding the same number to both terms to get an equivalent ratio ($2:3\neq 4:5$); equivalent ratios are got by multiplying or dividing.
- Comparing ratios by only looking at the first terms; convert to fractions with the same denominator.
- Mixing up which quantities go together in the unitary method (e.g. dividing cost by the wrong number of items).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Proportional Reasoning-1** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** ₹ 1000	Convert the reciprocals to whole-number ratio using the LCM
2	**(c)** 12 days	Find the work left in man-days and divide by the new number of men
3	**(c)** 7 : 3	Time is inversely proportional to speed
4	**(b)** 27	Adding the denominators keeps the ratio
5	**(b)** 9 : 25	Area $=$ side^2
6	**(b)** 15	$x = \frac{5\times18}{6}$
7	**(d)** 15 : 18 : 8	Make the middle terms equal using their LCM
8	**(d)** 20 years	Let the ages be $4x$ and $3x$; solving $(4x+5)\times4 = (3x+5)\times5$ gives $x = 5$
9	**(c)** A is true but R is false.	Cost of 1 pen $= 15$, so 10 pens cost 150 and A is true. The unitary method first finds the value of **one** unit, so R is false.
10	**(d)** A is false but R is true.	$\frac{2}{3} = \frac{8}{12}$ and $\frac{3}{4} = \frac{9}{12}$, so $2:3 < 3:4$; A is false, R is a true method.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Take $x = 9k$, $y = 5k$ (1); $(14k) : (4k) = 7 : 2$ (1).	1+1
12	Ratio $\frac{1}{3} : \frac{1}{4} = 4 : 3$ (1); parts ₹ 800 and ₹ 600 (1).	1+1
13	Perimeters $6 : 11$ (1); areas $36 : 121$ (1).	1+1
14	Take $x = 11k$, $y = 4k$ (1); $(15k) : (7k) = 15 : 7$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let ages be $4x$ and x (1); $(4x+10)\times2 = (x+10)\times5$ gives $x = 10$ (1); ages 40 and 10 years (1).	1+1+1
16	Total work $= 288$ man-days; done $= 96$ (1); left $= 192$ (1); days $= 192 \div 24 = 8$ (1).	1+1+1
17	LCM of 3, 5, 6 is 30; ratio $10 : 6 : 5$ (2); parts ₹ 1000, ₹ 600, ₹ 500 (1).	2+1
	OR: Let ages be $7x$ and $3x$ (1); $(7x+2)\times7 = (3x+2)\times15$ gives $x = 4$ (1); ages 28 and 12 years (1).	1+1+1
18	Total work $= 360$ man-days; done $= 72$ (1); left $= 288$ (1); days $= 288 \div 24 = 12$ (1).	1+1+1
	OR: LCM of 2, 5, 10 is 10; ratio $5 : 2 : 1$ (2); parts ₹ 2350, ₹ 940, ₹ 470 (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $4500 \div 6 =$ ₹750 (1). (ii) $750 \times 10 =$ ₹7,500 (1). (iii) $12000 \div 750 = 16$ days (1). (iv) Second mason earns $3900 \div 5 =$ ₹780 per day (1), which is ₹30 more per day than the first (1).	1+1+1+2
20	Investment × time: 24, 20, 24 (1); ratio $6 : 5 : 6$ (1); one part ₹ 80 (1); shares ₹ 480, ₹ 400, ₹ 480 (2).	1+1+1+2
	OR: Milk 70 L, water 30 L (1); in 10 L removed: milk 7 L, water 3 L (1); milk left 63 L (1); water now $30 - 3 + 10 = 37$ L (1); ratio $63 : 37$ (1).	1+1+1+1+1

Mistakes to avoid:

- Forming a ratio without converting units: 50 cm : 2 m is $50 : 200 = 1 : 4$, not $25 : 1$.
- Writing the ratio in the wrong order: the ratio of A to B is not the same as B to A.
- Dividing a total in the ratio $3 : 5$ by dividing it by 3 and 5 separately; first add the parts ($3 + 5 = 8$) and divide by 8.
- Adding the same number to both terms to get an equivalent ratio ($2 : 3 \neq 4 : 5$); equivalent ratios are got by multiplying or dividing.
- Comparing ratios by only looking at the first terms; convert to fractions with the same denominator.
- Mixing up which quantities go together in the unitary method (e.g. dividing cost by the wrong number of items).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Proportional Reasoning-1** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 9 : 5	Time is inversely proportional to speed
2	**(d)** 55	Adding the denominators keeps the ratio
3	**(d)** 16 : 49	Area $=$ side^2
4	**(a)** 24	$x = \frac{8\times15}{5}$
5	**(b)** 4 : 10 : 3	Make the middle terms equal using their LCM
6	**(d)** 21 years	Let the ages be $7x$ and $5x$; solving $(7x+3)\times3 = (5x+3)\times4$ gives $x = 3$
7	**(a)** ₹ 1175	Convert the reciprocals to whole-number ratio using the LCM
8	**(b)** 16 days	Find the work left in man-days and divide by the new number of men
9	**(a)** Both A and R are true and R is the correct explanation of A.	$\frac{5}{8}\times1200 = 750$; R gives the formula used.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	$40 : 120 = 1 : 3$ so A is true. R is true but only states that ratios are unitless; it does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Ratio $\frac{1}{2} : \frac{1}{3} = 3 : 2$ (1); parts ₹ 900 and ₹ 600 (1).	1+1
12	Perimeters $2 : 9$ (1); areas $4 : 81$ (1).	1+1
13	Take $x = 13k$, $y = 7k$ (1); $(20k) : (6k) = 10 : 3$ (1).	1+1
14	Ratio $\frac{1}{4} : \frac{1}{5} = 5 : 4$ (1); parts ₹ 1000 and ₹ 800 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let ages be $5x$ and $3x$ (1); $(5x+6)\times2 = (3x+6)\times3$ gives $x = 6$ (1); ages 30 and 18 years (1).	1+1+1
16	Total work $= 300$ man-days; done $= 100$ (1); left $= 200$ (1); days $= 200 \div 25 = 8$ (1).	1+1+1
17	LCM of 2, 3, 6 is 6; ratio $3 : 2 : 1$ (2); parts ₹ 1200, ₹ 800, ₹ 400 (1).	2+1
	OR: Let ages be $8x$ and $3x$ (1); $(8x+4)\times4 = (3x+4)\times9$ gives $x = 4$ (1); ages 32 and 12 years (1).	1+1+1
18	Total work $= 288$ man-days; done $= 72$ (1); left $= 216$ (1); days $= 216 \div 18 = 12$ (1).	1+1+1
	OR: LCM of 2, 4, 5 is 20; ratio $10 : 5 : 4$ (2); parts ₹ 1500, ₹ 750, ₹ 600 (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $2+3+5 = 10$ parts (1). (ii) $\frac{2}{10}\times15000 =$ ₹3,000 (1). (iii) $\frac{5}{10}\times15000 =$ ₹7,500 (1). (iv) B gets $\frac{3}{10}\times15000 =$ ₹4,500 (1); for ₹24,000, B gets $\frac{3}{10}\times24000 =$ ₹7,200 (1).	1+1+1+2
20	Investment × time: 24, 20, 12 (1); ratio $6 : 5 : 3$ (1); one part ₹ 110 (1); shares ₹ 660, ₹ 550, ₹ 330 (2).	1+1+1+2
	OR: Milk 75 L, water 45 L (1); in 24 L removed: milk 15 L, water 9 L (1); milk left 60 L (1); water now $45-9+24 = 60$ L (1); ratio $1 : 1$ (1).	1+1+1+1+1

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